BRK - Feedline

User manual and parameter reference · 1.3.1

A stereo delay whose feedback loop can contain up to ten processing modules.

Quick start

1. Insert Feedline on an audio track and play a short source.
2. Start with modest feedback and a delay time that separates the repeats.
3. Add a filter or other module to the loop, then move the output/feedback taps to compare the paths. Use Kill if you want to stop the repeats.

Editor window

1240 × 689 fixed. Dimensions are logical editor pixels, before host window chrome. HELP adds 92 pixels of height (Gear adds a one-pixel divider too). Allow enough usable screen space at your display scale. Press HELP again to close the help area.

Operation

The feedback tap determines what returns to the loop; the output tap determines what you hear. Module order therefore changes subsequent repeats, not just the current output.

Feedback above unity is intentional and can sustain a loop. The limiter contains the loop level, but choose your listening level deliberately. Kill, Kill on stop and the selected Kill style govern how repeats stop.

HELP follows named controls. The loop graph, EQ and modulator drawings still use the pointer. The complete module chain, routing and modulation are kept in presets and sessions.

The parameter table describes the initial module layout. Each of the ten slots reuses its host controls when its module changes; names, ranges and meaning then follow the selected module. Unused controls are reserved. The Modules section describes each available module.

The loop

`IN -> (+) -> DELAY -> module 1 ... module N`, with two markers on the line after the delay:

- **FB** is where the return leaves. From there it goes along the return lane: the limiter (Ceiling, Release), Feedback, Cross-feed (Stereo and Tape), the mode's routing, and back into `(+)`, where it joins the input and is written into the delay. Feedback goes to 120%, or to 200% with the small `200%` switch above it; with the switch off, modulation cannot push it past 120% either. The `Ø` switch beside it flips the polarity of the return: at short times the comb's peaks become notches (a hollower, odd-harmonic sound); at long times

every other repeat is upside down, which you hear mostly against the dry. The lane's knobs sit on the wire and are not selected; they are just turned.

- **OUT** is where you listen. From there the signal goes up and across to a column of Mix, Width and Output knobs and the OUT jack, on the right edge. The two markers move independently.

So the one chain does several jobs, depending on where the markers sit:

- Modules before both markers run on every pass and are heard: echo n has been through them n times.
- OUT after FB: modules between them run once, on the way out, and never go back into the loop.
- OUT before FB: modules between them are in the loop but past where you listen, so the first echo is clean of them and you first hear them on the second echo.
- Modules past both markers are not processed; it is a place to park them.

A module card shows its zone with a coloured bar (amber in the loop, dimmer amber in the loop past OUT, teal after the loop), and the zones are labelled behind the cards. The card draws what the module is doing with its values right now, modulation included (a filter's response, a drive's curve, a crushed sine, the stereo fan, turning reels with the tape wavering), with the values in words underneath and a dot per modulator on it beside the bypass dot.

Consequences of the order:

- Feedback is applied after the limiter, so it never changes the level of the first echo.
- The limiter caps what goes back into the delay (stereo-linked peak limiting, instant attack, no lookahead so it adds no loop latency, with a 5 Hz DC blocker in front). Above 100% feedback the repeats grow until they meet the ceiling and hold there. It is not on the output path: what you hear after FB is not limited.
- When FB and OUT sit together, you hear exactly what goes back into the loop, before the limiter.
- When OUT is before FB and a module between them has latency, you hear the echoes that much early; the loop itself stays on time.

Kill

`KILL` (top left of the loop) silences the repeats: the return and the echoes you hear. It is a move for breaks, not a panic button: the dry signal carries on. Releasing it empties the loop, even half way through a fade: it ducks for 5 ms, then everything the delay and the modules held is gone, so only what you play from then on repeats (raising Time afterwards does not reach the old audio either). It is a host parameter, so it can be automated. Click it to latch; hold it to kill only while held. The Kill controls sit in two rows: KILL and its style, then On stop and the kill time (a value box: drag it up or down).

- **Cut**: at once (5 ms).
- **Fade**: over Kill time (10 ms to 4 s).
- **Stop**: a tape stop. The heads slow to a halt over Kill time, so the repeats fall in pitch as they fade. On release the heads jump back to their place while the loop is emptied.

Delay modes

All four are the same engine: a stereo delay line, up to three read heads, and a routing for the return. Everything about the delay is on its card in the loop, which is not selected but turned where it sits: the mode selector (icons; hover one for its name), Time with its ms / note switch, When Time moves and Response, the mode's own controls between them, and the delay line itself underneath. On tape the own controls are a row per head: on, level and pan.

Mode	Read heads	Return	Own controls
Stereo	L at Time, R at Time x (1 + R offset)	Straight; Cross-feed (100% swaps the sides every pass)	R offset
Ping-pong	L and R at Time	Crossed. The input is summed to mono and enters on First side	First side
Shuffle	First side at Time x (1 + Swing), the other at Time x (1 - Swing)	Crossed, as ping-pong; each pair adds up to 2 x Time	Swing, First side
Tape	Three heads at 1x, 2x and 3x Time (the RE-201's ratios)	Straight, from every head that is on; Cross-feed	Per head: on, level, pan

- Time is a note value (synced to the host tempo) or milliseconds (5 ms to 2 s); synced, the card also shows what the note comes to in ms. It is one knob and one host parameter in both units: the switch only changes how it reads (and, synced, it steps from note to note), so automation and modulation stay on it. Switching from the editor keeps the time as near as it can (250 ms becomes 1/8 at 120 BPM, and back); switching by automation does not convert. In Tape mode it sets head 1, the way the RE-201's Repeat Rate sets tape speed. A modulator on a synced Time moves it smoothly through the note values, not in steps. Response also smooths modulation, so a fast LFO needs a short Response to be heard in full.
- **When Time moves** (top right of the delay card, for every mode): `Glide` ramps the read speed (pitch between 0.1x and 1.9x), so the pitch bends like tape; `Crossfade` moves to the new time with an equal-power crossfade and no pitch change. Response (5 ms to 4 s) sets how fast: the glide's time constant (short is a quick jump, long a slow bend), or the crossfade's length.
- A mode change fades the heads out over 15 ms, switches, and fades them back in.
- Tape mode has no tape character of its own. Wow, flutter and saturation are the Tape module.
- **On stop** (under KILL, off by default): while the host is stopped, Kill holds the repeats, in its style and time, so stopping fades or tape-stops the loop and play starts clean; the KILL button glows dimly meanwhile. Off, the repeats carry on when you stop.

Modules

The chain has 10 slots. + puts a module in a free slot and adds it at the end of the chain; a marker that was at the end stays at the end, so the new module lands in the zone the + sits in. Any type can be used any number of times. Drag a card to reorder (the markers keep their neighbours), click its dot to bypass it, right-click it to bypass or remove it. Every module has Mix and Bypass; a module bypassed all the way starts clean when it is turned back on, rather than playing what it held. Removing the selected module selects the one after it, or the one before it if it was the last.

Every slot has a fixed set of host parameters (16 controls, Mix and Bypass), so every control of every module can be automated and modulated. The parameters take their names and units from the module in the slot ("3 EQ: 2 freq", "500 Hz") and tell the host when that changes. Reordering moves a module, not its slot, so automation follows it.

The modules are first versions, to be designed one at a time:

- **Filter**: low cut and high cut, each with its own resonance, at 12 or 24 dB an octave (at 24 dB a flat stage comes first and the resonant one second). Spread (-1 to +1 octave) moves the right side's cutoffs up and the left's down by that much (the other way below zero), so in the loop the echoes pull further apart with each pass; at a full octave the sides' correlation on noise falls to about 0.2. The card draws the right side fainter.
- **Band**: one band-pass (0 dB at its peak), notch or peak (with Gain); Frequency and Resonance (Q 0.5 to 12).
- **EQ**: four bands, each off, a low or high cut, a low or high shelf, or a bell (frequency, gain, Q). Its panel draws the response with a node per band: drag a node for frequency and gain, use the wheel over it for Q.
- **Drive**: a blend of soft and hard clipping (Hardness), Asymmetry for even harmonics. It adds loop gain.
- **Crush**: bit depth and sample-and-hold rate reduction (Downsample divides 48 kHz at any sample rate, so it sounds the same at every rate and in the echo view), and a noise floor added after it: White, Pink, Crackle (sparse clicks, like dust) or Hum (50 Hz and odd harmonics), Off up to -18 dB (the noise's RMS; crackle's clicks peak there), with Width from one source in both sides (mono) to a separate one per side (for hum, the right side's phase moves up to a quarter turn). In the loop the noise builds up with the feedback, like a worn machine's floor.
- **Tape**: wow, flutter, saturation and a high cut (Tone), with a small dial beside Wow and Flutter for their speed (Wow 0.1 to 4 Hz, 0.55 to start; Flutter 2 to 20 Hz, 6.7 to start). 4 ms of latency.
- **Diffuse**: six allpass stages per side; Amount and Size.
- **Shift**, in three modes (the controls that do not apply are hidden); Spread takes R the other way in each:
    - Frequency: a frequency shifter (the shared `FrequencyShifterDSP`). Shift is a share (+/-100%, finer near zero) of Range (1 Hz to 20 kHz), so a small Range gives fractions of a hertz. 31 samples of latency.
    - Pitch: -24 to +24 semitones, from two grains of a 50 ms window crossfaded at equal power; +12 in the loop is the shimmer. 25 ms of latency.
    - Ring: ring modulation by a Carrier (1 Hz to 5 kHz), made up by 3 dB so a ring loop decays like any other. Spread puts the right carrier up to a quarter turn ahead: a turning pan at low rates, sidebands apart in phase at audio rates. At full Spread the sides are uncorrelated at equal level, and the mono sum is 3 dB down, never cancelled.
- **Phaser**: six allpass stages swept around Centre by Rate and Depth (up to two octaves each way), with Feedback, summed with the input. The right side's sweep runs a quarter turn ahead.
- **Flanger**: a 0.5 to 10 ms delay swept by Rate and Depth, with Feedback (either polarity), summed with the input.
- **Chorus**: two voices a side around a 5 to 30 ms delay, swept in opposition, summed with the input.

- **Gate**: a downward expander: below Threshold the level falls Ratio times faster (1:1 to Gate), down to Range; both sides linked. The level detector catches peaks at once and holds them 20 ms; the gate opens over Attack and closes over Release. Its card draws the gain against the input level, with Range as a faint floor. In the loop the tails fall away sooner and the gaps stay clean.

- **Transient**: a transient shaper, the same at any level. The level follows peaks at once and holds them 15 ms (so a low note's cycles do not flutter it), then falls in 20 ms. Attack (-100% to +100%) takes how far the level stands above a copy of it that rises only over Attack time (1 to 100 ms, 15 to start), times the amount, as gain: +100% doubles the start of a hit in dB, -100% flattens it to the body. Sustain does the same with a copy that falls only over Sustain time (20 ms to 2 s, 200 to start), in the tail. Each distance counts up to 24 dB, and below -60 dB the gain lets go (gone at -80 dB), so the noise floor is not pumped; a steady sound passes untouched. Link at 100% gives both sides the louder side's gain, so the image holds; at 0% each side is shaped on its own (a ping-pong's repeats on their own side). A new one starts at Attack +50%, Sustain -30%. Its card draws a hit (a click falling into a slower body, over 600 ms) faint, and what the module makes of it. In the loop every repeat is shaped again: Sustain down turns the echoes into clicks, Attack up keeps them sharp while filters dull them.

- **Reverb**: a diffuser into an eight-line feedback delay network. The diffuser spreads the input over eight channels and smears it in four steps (delay each channel, shuffle with some flipped, Hadamard mix); the network's lines feed each other through a Householder mix, each with a damping low-pass and a slow wobble in length. Size sets the room (12 to 100 ms; the lines run 0.6 to 1.2 rooms), Decay the time to fall 60 dB (0.2 to 12 s; it measures about 10% short at 6 kHz Damping), Damping darkens it, Pre-delay up to 100 ms. Diffusion fades from the plain input into the lines (sparse, distinct first reflections) to the smeared one (a smooth bloom that is noise-like within about 30 ms); Motion wobbles the lines by up to 2 ms so the tail does not ring. Only the reverb comes out, at about -4 dB under the input at 2 s; a new one starts at Mix 35%, since at full it would wash out the echoes. In the loop every echo is reverberated again.

- **Delay**: a delay with its own Feedback, in ms (1 ms to 2 s) or synced to the host's tempo (Notes: the main Time's note values, 1/8D to start), up to 8 s. Its Time is one knob for both units, as on the delay card. Only the delayed signal comes out, so in the loop it adds its time and repeats to the loop's (polyrhythms, multi-taps); Mix brings the input back, and a new one starts at 50%. Stereo: R offset makes the right side's time up to 50% longer or shorter. Ping-pong: the input goes in mono on the left and every repeat crosses sides. It reads its line with the same cubic interpolation as the loop (Flanger and Chorus too), which at a half-sample delay loses about 0.2 dB a repeat at 8 kHz and 1 dB at 12 kHz.

- **Width**: rotation and mid/side width (-200% to 200%), applied on every pass when in the loop. 0% is mono; below 0% the sides swap (-100% is a straight L/R flip), so in the loop the repeats alternate between mirrored and not. The card draws the field across: L as a dot, R as a ring, the speakers as ticks.

- **Spectral**: one short-time FFT (square-root Hann, quarter-frame hop) and six effects on the same bins, in this order. All of them together cost one frame of latency, the FFT Size (512, 1k or 2k at 48 kHz: 11, 21 or 43 ms, and 1k to start; the frame scales with the sample rate, so it lasts as long and a bin is as many hertz wide at any rate, including the echo view's 24 kHz). A bin is 94, 47 or 23 Hz wide, so the smaller sizes resolve less in the lows.

    - **Pitch**: -24 to +24 semitones, a phase vocoder with its phases locked to the peaks: a partial lands on the exact pitch, and since a peak moves by whole bins its level can dip up to about 3 dB, most at the small sizes.
    - **Blur** lets each bin's level follow slowly (30 ms to 3 s); **Freeze** holds the spectrum, its frequencies (averaged over the last few frames, so a sound that stops as Freeze comes on is held in tune) and the shape of its phases. Turn Feedback down while frozen, or the held sound builds up in the loop.

After the loop is emptied (Kill) or the Size changes, Freeze waits for a frame of new input before it takes its hold.

- **Scatter**: a fixed random phase per bin, a different set per side: diffusion and width.
- **Tilt** only cuts (darker: above Pivot, brighter: below it), so it can never add loop gain.
- **Gate**: bins below Threshold fall 3 dB per dB under, down to Range; they open at once and close over 80 ms.
- **Suppress**: a bin that stands more than 3 dB above the average of its neighbourhood (in dB, over the level of the last 30 ms, an octave and a half to a sixth wide by Sharpness, half an octave at 50%) is a resonance, and Depth of its excess is taken away (at most 30 dB), between Low and High. Speed sets the release (400 ms to 20 ms; the attack is a quarter of it). On white noise it takes less than 0.1 dB; a sine 40 dB over the noise loses 17 dB at full Depth and 50% Sharpness at 1k, 25 dB at 2k.

Without Pitch, Blur or Freeze the bins keep their own phases, and with every effect off the module is a clean delay of its Size. The Tilt, Gate and Suppress gains come from both sides' level together, so the image holds still. It mixes its own dry, delayed to match, and keeps running at Mix 0 and when bypassed, so the loop's timing does not move. The card draws the level going in and, hanging from the top, the gain each band gets on this pass.

Phaser, Flanger and Chorus sum their wet path with the input at half each, so at Mix 100% they are the effect itself and the level holds in the loop.

The module panel lays a module's controls out in titled groups (Low and High for the Filter, Motion and Tone for the Phaser…), with a stepped choice such as a mode or a slope as a row of cells.

Latency of modules before FB is taken off the delay time, so echoes stay on time. A module's latency can follow its settings (Shift's mode, Spectral's Size), so it is summed again every block. It counts whether the module is on or not: a module's dry path is delayed by its latency, so Bypass and Mix never move the timing. Latency after FB only delays what you hear. The delay can then be no shorter than the loop's latency plus 8 samples: when that is longer than Time's own 5 ms, the delay card shows it under the time switch ("min 22 ms"), in red while a head is held there.

Modulators

8 modulator slots, each with 4 host parameters, named after what the slot holds. A slot is one of:

- **LFO**: Rate, Shape (sine, triangle, saw up, saw down, square, random) and Phase. A switch beside Rate picks its unit: Hz (0.02 to 20 Hz) or a note value (1/16 to 4 bars, following the song while the host plays); the note values are the slot's Sync parameter, whose 0 is Free. It swings each target both ways around the target's own value.
  - **Stopped** (Rate at the bottom of the knob, in Hz): the LFO is a macro. Phase, now called Position, moves its output in a straight line from -100% to +100% whatever the shape (Shape is hidden), 0 at the centre: its targets go down to the left of centre and up to the right. A stopped LFO counts as static in the echo view and the knobs' value text, like an envelope.
- **Envelope**: follows the input or the echoes (Source), with Attack, Release and Gain.

A slot that drives nothing is off: its name is dimmed and its meter hidden, though it keeps running so its scope shows what it would do. Each shows a scope of its output over the last three seconds beside its controls.

Any continuous control can be a target: module controls and mixes, the core's controls (Time, R offset, Cross-feed, Swing, head levels and pans, Feedback, the limiter, Mix, Width, Output, Kill time), and the modulators' own controls. A target keeps its own value, which you can still turn; every modulator on it adds its amount on top, and one control can take several. With every stopped LFO at its centre and no running LFOs or envelopes mapped, the knobs are exactly as set.

Mapping is the same for every kind: click a modulator's `MAP`, then drag any knob up or down (the drag sets the amount, from the knob's value) or click it (all the way, or for an LFO as far as it can swing both ways; click again to unmap). Click `MAP` again to finish. A modulated knob draws each modulator as an outer arc in its colour, with a dot where the control is now, and its value text is what you hear. Right-click a knob to remove a modulator from it. Double-click a modulator's name to rename it.

A new instance comes with three stopped LFOs, mapped and at their centre (Darken: Filter high cut down to the right; Wear: Tape wow and saturation up; Throw: Feedback from 55% up to 110%, down to 0% to the left) and five running LFOs mapped to nothing, the first named Drift.

Presets

The header holds the preset bar: previous and next, the preset's name (a star marks a patch that has changed since it was loaded, `newer` one from a preset made with a newer version, whose unknown settings were skipped; click it for the list, with your folders as submenus, Init, Save as, Rename, Delete and the presets folder) and Save (`Folder/Name` saves into a folder; saving over another preset asks first). Dropping a `.feedline` file on the bar loads it. A preset is the whole patch: every parameter but Output and Kill, the chain (which module is in which slot, the order, the markers) and the modulators (kinds, names and routings). Loading one sets every parameter it covers once, to the file's value or its default, so a preset saved before a control existed loads with that control at rest. Init is a new instance's patch. Your presets are `.feedline` files in `~/Library/Audio/Presets/Barakigal Audio/Feedline`. Factory presets are `feedline/presets/*.feedline` in the repository (numbered, so they list in order), built into the plugin; there are none yet. A session remembers the preset it came from (by its path in the presets folder) and whether it had changed.

Presets and sessions store each choice parameter (Mode, First side, When Time moves, Kill style) and each modulator's kind by a stable ID beside its position, and read the ID first (the position only from a file saved before the IDs), so a list can change without an older file meaning another entry. An ID never changes once released. These choices are not automatable by the host, which would store their place in the list; the on/off switches are toggles and stay automatable. The modules' and modulators' own choices (a filter's Slope, an LFO's Shape) are still stored as a fraction of their slot's generic 0 to 1 parameter, so changing one of those lists, even at its end, needs an upgrade step for older files, and moves automation recorded on that slot.

The editor

- **Echo view** (top): one pluck sent through a private copy of the engine on a background thread whenever a setting changes, so it shows what the loop does to a hit: L above, R below, on a 42 dB scale, with the limiter ceiling.
- **Loop view**: every wire draws the signal passing through it, with trails, and is brighter where it is louder. The delay card shows the delay line itself, newest audio on the left moving right to the read heads. The return lane holds the limiter (Ceiling, Release and its gain reduction), Feedback, Cross-feed and the mode's routing (the strands cross for ping-pong and shuffle, and partly cross with Cross-feed).

- **Bottom**: two panels, half the width each. The left one holds the controls of the module you click (its Mix, then Bypass and Remove as icon cells, sit at the right of its header, in the same place for every module; the longer explanation is the panel's tooltip). A module is always selected while there are any. The right one, without a header, holds the modulators: the eight slots as a strip of cells, with live meters and `MAP` buttons (a slot's tooltip lists what it drives), under it a strip with the selected one's name (double-click to rename) and its kind, then its controls. Everything else (the delay card, Kill, the limiter, Feedback and its switches, Cross-feed, Mix, Width, Output) is turned where it sits on the drawing.

Every parameter

Values and defaults are taken from the plugin. A control's effect can depend on the selected mode; see the operation notes above. Choices keep their saved identity between versions. Knobs and switches can be automated unless marked otherwise.

Parameter	What it does	Values and default
Mode	Tape: One tape and three playback heads at 1x, 2x and 3x Time, as on the RE-201. Time sets head 1. Heads that are on feed both the output and the loop; Cross-feed, on the return, bounces the repeats between the sides.	Stereo, Ping-pong, Shuffle, Tape Default: Ping-pong Not host automatable
Sync	Notes: Time in note values, following the host's tempo.	Off to On Default: On
Time	Kill time: how long Fade and Stop take. Drag up or down.	1/32 to 1 bar Default: 1/8
R Offset	Moves the right delay relative to the left as a share of Time; its interpretation depends on delay mode.	-50% to +50% Default: +12%
Cross-feed	Sends a share of each channel into the other side of the feedback loop.	0% to 100% Default: 0%
Swing	Offsets alternating taps in the applicable delay mode.	-50% to +50% Default: +22%
First Side	The side the input enters on; the repeats then alternate from there.	L, R Default: L Not host automatable
Time Change	Crossfade: a crossfade to the new time, no pitch change. Applies in every mode, and to Time moved by a modulator.	Glide, Crossfade Default: Glide Not host automatable
Time Response	Sets how quickly delay time follows an edit or modulation, according to Time Change mode.	5 ms to 4.00 s Default: 100 ms
Head 1	Enables this repeat tap at its numbered multiple of Time.	Off to On Default: On

Parameter	What it does	Values and default
Head 1 Level	Sets the signal level at this stage. Output is the final listening trim unless the signal-flow notes describe a wet-only output control.	0% to 100% Default: 100%
Head 1 Pan	Places or spreads the signal across the stereo field.	L100 to R100 Default: L50
Head 2	Enables this repeat tap at its numbered multiple of Time.	Off to On Default: Off
Head 2 Level	Sets the signal level at this stage. Output is the final listening trim unless the signal-flow notes describe a wet-only output control.	0% to 100% Default: 85%
Head 2 Pan	Places or spreads the signal across the stereo field.	L100 to R100 Default: R50
Head 3	Enables this repeat tap at its numbered multiple of Time.	Off to On Default: On
Head 3 Level	Sets the signal level at this stage. Output is the final listening trim unless the signal-flow notes describe a wet-only output control.	0% to 100% Default: 75%
Head 3 Pan	Places or spreads the signal across the stereo field.	L100 to R100 Default: C
Feedback	Modulated by Throw. Right-click to remove.	0% to 200% Default: 55%
Feedback to 200%	Returns part of a signal to its input. More feedback increases the strength or persistence of the selected effect.	Off to On Default: Off
Feedback Invert	Returns part of a signal to its input. More feedback increases the strength or persistence of the selected effect.	Off to On Default: Off
Limiter Ceiling	Sets the peak ceiling in the feedback loop.	-24.0 dB to 0.0 dB Default: -6.0 dB
Limiter Release	Sets the fall after release or the recovery time of the selected processor; see the envelope or detector description above.	10 ms to 600 ms Default: 140 ms
Mix	Blends the processed and dry signals. The signal-flow notes specify whether Output changes the wet signal alone or the complete blend.	0% to 100% Default: 40%
Width	Places or spreads the signal across the stereo field.	0% to 150% Default: 100%
Output	Sets the signal level at this stage. Output is the final listening trim unless the signal-flow notes describe a wet-only output control.	-24.0 dB to +12.0 dB Default: 0.0 dB
Kill	Kill the repeats. Click to latch, hold to kill only while held.	Off to On Default: Off

Parameter	What it does	Values and default
Kill Style	Stop: the tape slows to a halt over the kill time.	Cut, Fade, Tape stop Default: Fade Not host automatable
Kill Time	Sets how long the chosen Kill action takes.	10 ms to 4.00 s Default: 300 ms
Kill on Stop	Runs Kill when host playback stops.	Off to On Default: Off
1 Filter: Low cut	Sets the high-pass boundary of this Filter module.	20 Hz to 2.0 kHz Default: 200 Hz
1 Filter: High cut	Sets the low-pass boundary of this Filter module.	500 Hz to 20 kHz Default: 3.2 kHz
1 Filter: High res	Sets resonance at the high-cut boundary.	0% to 100% Default: 50%
1 Filter: Low res	Sets resonance at the low-cut boundary.	0% to 100% Default: 50%
1 Filter: Slope	Chooses the filter circuit, response or slope. Variants depend on the selected model; see the filter operation notes.	12 dB to 24 dB Default: 24 dB
1 Filter: Spread	Separates the left and right filter frequencies.	L +1.00 oct to R +1.00 oct Default: Off
1 Filter: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
1 Filter: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
1 Filter: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
1 Filter: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
1 Filter: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
1 Filter: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%

Parameter	What it does	Values and default
1 Filter: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
1 Filter: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
1 Filter: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
1 Filter: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
1 Filter: Mix	Blends the processed and dry signals. The signal-flow notes specify whether Output changes the wet signal alone or the complete blend.	0% to 100% Default: 100%
1 Filter: Bypass	Switches between processing and the dry signal. It does not start Auto Gain learning.	Off to On Default: Off
2 Tape: Wow	Sets the depth of slow tape-speed variation.	0% to 100% Default: 50%
2 Tape: Flutter	Sets the depth of faster tape-speed variation.	0% to 100% Default: 50%
2 Tape: Saturation	Adds tape-style nonlinear coloration.	0% to 100% Default: 50%
2 Tape: Tone	Sets the frequency boundary of this filter or band. Its mode and slope determine which frequencies it changes.	1.5 kHz to 18 kHz Default: 5.2 kHz
2 Tape: Wow speed	Sets the rate of the slow tape-speed variation.	0.10 Hz to 4.00 Hz Default: 0.63 Hz
2 Tape: Flutter speed	Sets the rate of the faster tape-speed variation.	2.00 Hz to 20.0 Hz Default: 6.32 Hz
2 Tape: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
2 Tape: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
2 Tape: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%

Parameter	What it does	Values and default
2 Tape: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
2 Tape: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
2 Tape: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
2 Tape: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
2 Tape: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
2 Tape: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
2 Tape: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
2 Tape: Mix	Blends the processed and dry signals. The signal-flow notes specify whether Output changes the wet signal alone or the complete blend.	0% to 100% Default: 100%
2 Tape: Bypass	Switches between processing and the dry signal. It does not start Auto Gain learning.	Off to On Default: Off
3 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
3 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
3 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
3 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%

Parameter	What it does	Values and default
3 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
3 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
3 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
3 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
3 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
3 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
3 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
3 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
3 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
3 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
3 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
3 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
3 Empty: Mix	Blends the processed and dry signals. The signal-flow notes specify whether Output changes the wet signal alone or the complete blend.	0% to 100% Default: 100%

Parameter	What it does	Values and default
3 Empty: Bypass	Switches between processing and the dry signal. It does not start Auto Gain learning.	Off to On Default: Off
4 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
4 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
4 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
4 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
4 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
4 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
4 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
4 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
4 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
4 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
4 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
4 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%

Parameter	What it does	Values and default
4 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
4 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
4 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
4 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
4 Empty: Mix	Blends the processed and dry signals. The signal-flow notes specify whether Output changes the wet signal alone or the complete blend.	0% to 100% Default: 100%
4 Empty: Bypass	Switches between processing and the dry signal. It does not start Auto Gain learning.	Off to On Default: Off
5 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
5 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
5 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
5 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
5 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
5 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
5 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%

Parameter	What it does	Values and default
5 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
5 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
5 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
5 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
5 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
5 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
5 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
5 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
5 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
5 Empty: Mix	Blends the processed and dry signals. The signal-flow notes specify whether Output changes the wet signal alone or the complete blend.	0% to 100% Default: 100%
5 Empty: Bypass	Switches between processing and the dry signal. It does not start Auto Gain learning.	Off to On Default: Off
6 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
6 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%

Parameter	What it does	Values and default
6 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
6 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
6 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
6 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
6 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
6 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
6 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
6 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
6 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
6 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
6 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
6 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
6 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
6 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%

Parameter	What it does	Values and default
6 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
6 Empty: Mix	Blends the processed and dry signals. The signal-flow notes specify whether Output changes the wet signal alone or the complete blend.	0% to 100% Default: 100%
6 Empty: Bypass	Switches between processing and the dry signal. It does not start Auto Gain learning.	Off to On Default: Off
7 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
7 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
7 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
7 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
7 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
7 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
7 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
7 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
7 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
7 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%

Parameter	What it does	Values and default
7 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
7 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
7 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
7 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
7 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
7 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
7 Empty: Mix	Blends the processed and dry signals. The signal-flow notes specify whether Output changes the wet signal alone or the complete blend.	0% to 100% Default: 100%
7 Empty: Bypass	Switches between processing and the dry signal. It does not start Auto Gain learning.	Off to On Default: Off
8 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
8 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
8 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
8 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
8 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%

Parameter	What it does	Values and default
8 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
8 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
8 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
8 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
8 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
8 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
8 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
8 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
8 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
8 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
8 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
8 Empty: Mix	Blends the processed and dry signals. The signal-flow notes specify whether Output changes the wet signal alone or the complete blend.	0% to 100% Default: 100%
8 Empty: Bypass	Switches between processing and the dry signal. It does not start Auto Gain learning.	Off to On Default: Off

Parameter	What it does	Values and default
9 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
9 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
9 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
9 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
9 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
9 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
9 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
9 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
9 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
9 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
9 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
9 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
9 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
9 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%

Parameter	What it does	Values and default
9 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
9 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
9 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
9 Empty: Mix	Blends the processed and dry signals. The signal-flow notes specify whether Output changes the wet signal alone or the complete blend.	0% to 100% Default: 100%
9 Empty: Bypass	Switches between processing and the dry signal. It does not start Auto Gain learning.	Off to On Default: Off
10 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
10 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
10 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
10 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
10 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
10 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
10 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
10 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%

Parameter	What it does	Values and default
10 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
10 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
10 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
10 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
10 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
10 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
10 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
10 Empty: -	Reserved control with no audible role in the current module. Loading a different module can assign this slot a new control, range and name.	0% to 100% Default: 50%
10 Empty: Mix	Blends the processed and dry signals. The signal-flow notes specify whether Output changes the wet signal alone or the complete blend.	0% to 100% Default: 100%
10 Empty: Bypass	Switches between processing and the dry signal. It does not start Auto Gain learning.	Off to On Default: Off
M1 Darken: Rate	Sets the rate of this modulator in the units selected by its Sync mode.	Stopped to 20.0 Hz Default: Stopped
M1 Darken: Shape	Chooses or changes the source waveform or curve. Shape changes harmonic content; curve controls change how a segment progresses.	Sine to Random Default: Sine
M1 Darken: Sync	Switches this modulator between free time and musical timing.	Free to 4 bars Default: Free
M1 Darken: Position	Sets the play position of the selected modulation curve.	-100% to +100% Default: -100%

Parameter	What it does	Values and default
M2 Wear: Rate	Sets the rate of this modulator in the units selected by its Sync mode.	Stopped to 20.0 Hz Default: Stopped
M2 Wear: Shape	Chooses or changes the source waveform or curve. Shape changes harmonic content; curve controls change how a segment progresses.	Sine to Random Default: Sine
M2 Wear: Sync	Switches this modulator between free time and musical timing.	Free to 4 bars Default: Free
M2 Wear: Position	Sets the play position of the selected modulation curve.	-100% to +100% Default: -100%
M3 Throw: Rate	Sets the rate of this modulator in the units selected by its Sync mode.	Stopped to 20.0 Hz Default: Stopped
M3 Throw: Shape	Chooses or changes the source waveform or curve. Shape changes harmonic content; curve controls change how a segment progresses.	Sine to Random Default: Sine
M3 Throw: Sync	Switches this modulator between free time and musical timing.	Free to 4 bars Default: Free
M3 Throw: Position	Sets the play position of the selected modulation curve.	-100% to +100% Default: -100%
M4 Drift: Rate	Sets the rate of this modulator in the units selected by its Sync mode.	Stopped to 20.0 Hz Default: Stopped
M4 Drift: Shape	Chooses or changes the source waveform or curve. Shape changes harmonic content; curve controls change how a segment progresses.	Sine to Random Default: Sine
M4 Drift: Sync	Switches this modulator between free time and musical timing.	Free to 4 bars Default: Free
M4 Drift: Phase	Sets the phase or phase behavior described in the source and synchronization notes.	0° to 360° Default: 0°
M5 LFO: Rate	Sets the rate of this modulator in the units selected by its Sync mode.	Stopped to 20.0 Hz Default: Stopped
M5 LFO: Shape	Chooses or changes the source waveform or curve. Shape changes harmonic content; curve controls change how a segment progresses.	Sine to Random Default: Sine
M5 LFO: Sync	Switches this modulator between free time and musical timing.	Free to 4 bars Default: Free
M5 LFO: Phase	Sets the phase or phase behavior described in the source and synchronization notes.	0° to 360° Default: 0°

Parameter	What it does	Values and default
M6 LFO: Rate	Sets the rate of this modulator in the units selected by its Sync mode.	Stopped to 20.0 Hz Default: Stopped
M6 LFO: Shape	Chooses or changes the source waveform or curve. Shape changes harmonic content; curve controls change how a segment progresses.	Sine to Random Default: Sine
M6 LFO: Sync	Switches this modulator between free time and musical timing.	Free to 4 bars Default: Free
M6 LFO: Phase	Sets the phase or phase behavior described in the source and synchronization notes.	0° to 360° Default: 0°
M7 LFO: Rate	Sets the rate of this modulator in the units selected by its Sync mode.	Stopped to 20.0 Hz Default: Stopped
M7 LFO: Shape	Chooses or changes the source waveform or curve. Shape changes harmonic content; curve controls change how a segment progresses.	Sine to Random Default: Sine
M7 LFO: Sync	Switches this modulator between free time and musical timing.	Free to 4 bars Default: Free
M7 LFO: Phase	Sets the phase or phase behavior described in the source and synchronization notes.	0° to 360° Default: 0°
M8 LFO: Rate	Sets the rate of this modulator in the units selected by its Sync mode.	Stopped to 20.0 Hz Default: Stopped
M8 LFO: Shape	Chooses or changes the source waveform or curve. Shape changes harmonic content; curve controls change how a segment progresses.	Sine to Random Default: Sine
M8 LFO: Sync	Switches this modulator between free time and musical timing.	Free to 4 bars Default: Free
M8 LFO: Phase	Sets the phase or phase behavior described in the source and synchronization notes.	0° to 360° Default: 0°

Presets and files

Presets are browsed with the arrows or the preset-name menu. A star marks an edited patch. Save supports folders, Rename, Delete and opening the user preset folder; dropping a compatible preset file onto the bar loads it. A newer-file indication means this version could not apply every setting.

The host session stores the complete patch. A product preset can deliberately leave listening trim, audition or display controls unchanged; the product-specific rules below describe those exceptions.

Feedline presets use .feedline in ~/Library/Audio/Presets/Barakigal Audio/Feedline. Output and the momentary Kill state are left unchanged on preset loading. There is no bundled factory bank; save the loops you build.

Installation and help

Download the macOS archive from the product page and quit your audio host. Place the AU .component in ~/Library/Audio/Plug-Ins/Components, VST3 in ~/Library/Audio/Plug-Ins/VST3, or CLAP in ~/Library/Audio/Plug-Ins/CLAP. The product page also offers an installer with format selection. Restart or rescan the host. To uninstall, quit the host and remove that product's bundles from those folders; saved sessions and user presets are separate.

For questions or a problem report, use BRK Audio support on GitHub. Include the plugin version, macOS version, host and format, sample rate and a short description of how to reproduce it.